

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2023-2024

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 75

CSE 4501: Operating Systems**Programmable calculators are not allowed. Do not write anything on the question paper.**Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Race conditions are possible in many computer systems. Consider a banking system that maintains an account balance with two functions - deposit and withdraw. The amount of money to be deposited to or withdrawn from an account is passed via the parameter amount to each of these functions. Assume that a bank account with an initial balance of BDT 10000 is shared between two brothers. One of them attempts to withdraw an amount of BDT 4500 from the account while the other attempts to deposit an amount of BDT 5000 to the account, concurrently. Illustrate how a race condition is possible in this scenario. 3
(CO1)
(PO1)
- b) What three requirements must a solution to a critical-section problem satisfy? Devise an algorithm that fulfills these three requirements using the
`test_and_set (boolean *target)` 3 + 3
(CO1)
(PO1)
instruction. Provide a step-by-step illustration showing that your algorithm full fills all three requirements
- c) Consider *three* concurrent processes P_0, P_1 , and P_2 with statements S_0, S_1 , and S_2 , respectively. Now you are asked to propose an algorithm that blends the concepts of *Mutex Locks*, *Counting Semaphores* (CS), and *busy waiting* to ensure the execution of the statements in the order - $S_2 > S_1 > S_0$. Suppose your algorithm invokes the functions — `waitNacquire (pid)` and `signalNrelease ()` in the entry and exit sections of your code, respectively, while considering the global variables - CS and lock. The subscript i in the notation P_i indicates the process ID (pid). Based on this information, answer the followings -
 - i. Implement the two functions `waitNacquire (pid)` and `signalNrelease ()`. 3 + 3
(CO3)
(PO2)
 - ii. Using your implementation of these functions in Question 1.c)i, show the step-by-step execution of your algorithm with a proper illustration of the process timeline diagram. 5
(CO3)
(PO2)
2. a) With respect to the RPC mechanism, consider the *exactly once* semantic and answer the following questions —
 - i. Does the algorithm for implementing this semantic execute correctly even if the ACK message back to the client is lost due to a network problem? Describe the sequence of messages and discuss whether *exactly once* is still preserved. 5
(CO2)
(PO1)
 - ii. Assume that a distributed system is susceptible to server failure. What mechanisms would be required to guarantee the *exactly once* semantics for the execution of RPCs? 5
(CO2)
(PO1)
- b) What do you understand by *cascading termination*? What are the reasons for a parent process to terminate the execution of its child processes? Explain the statement, "All Orphan processes are Zombie processes but all Zombie processes are not Orphan processes." 2 + 3
(CO1)
(PO1)

- c) Briefly discuss the working mechanism of the message passing facility designed for the Windows operating system with appropriate illustration. 7
(CO1)
(PO1)
3. a) Briefly discuss the concepts of *Concurrency* and *Parallelism* with a suitable example and necessary diagram(s). 3
(CO1)
(PO1)
- b) Before threads became popular, the creation of multiple processes was commonly used to accomplish multiple tasks concurrently. Threading provides several benefits over process creation. A thread library provides the programmer with an API for creating and managing threads, and potentially includes the mechanism of task and data parallelism. Pthreads, the threads extension of the POSIX standard that defines an API for thread creation and synchronization, may be provided as either a user-level or a kernel-level library. While numerous systems implement the Pthreads specification, operating-system designers may implement it however they wish. Based on this information, answer the following questions —
- i. Discuss the benefits of multithreaded programming over the traditional *process creation* method 4
(CO1)
(PO1)
- ii. Consider a multithreaded program that evaluates the expression ' $(a+b)-(b/c)+(c \times d)$ ' on a system with $N = 5$ processing cores. Apply Amdahl's law to calculate the performance gain with this configuration. Mathematically show how the program's serial portion can disproportionately affect the performance gain with additional computing cores. 3 + 3
(CO2)
(PO2)
- iii. Write a multithreaded program using Pthreads that calculates various statistical values for a list of numbers. This program will be passed with a series of numbers on the command line and will then create *three* separate worker threads. One thread will determine the *average* of the numbers, the second will determine the *maximum* value, and the third will determine the *minimum* value. The variables representing the *average*, *minimum*, and *maximum* values will be stored globally. The worker threads will set these values, and the parent thread will output the values once all the workers have exited. 12
(CO3)
(PO2)
- For example, if you pass the integers 90 81 78 95 79 72 85 to your program, it will report the following —
- The average value is 82
The minimum value is 72
The maximum value is 95
- [N.B.: Your answer must include full implementation of all the functions.]

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MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4503: Microprocessor and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
- | | | |
|----|---|-------------------------|
| 1. | a) Define and explain differences between the STACK memory operations of 8085 and 8086 microprocessors. | 10
(CO1)
(PO1) |
| | b) Write detailed executions of any arithmetic instructions each for 8085 and 8086 microprocessors, respectively; where, use of <i>Accumulator</i> register is a must. | 10
(CO2)
(PO2) |
| | c) Explain the concept of <i>Label</i> and <i>Loop</i> in assembly language programming. | 5
(CO1)
(PO1) |
| 2. | a) Differentiate I/O operations for BIOS and DOS routines with proper explanation. | 10
(CO2)
(PO1) |
| | b) Derive the corresponding machine codes for the following instructions using their respective instruction format and also show how the machine codes of the instructions are to be stored in memory:
i. IN DX, AB12H
ii. MOV BX, 01010101B
iii. MOV CS: [BP + F1E2H], DL | 3 × 3
(CO1)
(PO2) |
| | c) Using appropriate example codes, write the alternative details of the following assembly language instructions:
i. XCHG
ii. RET. | 2 × 3
(CO4)
(PO2) |
| 3. | a) Suppose, Instruction Pointer (IP) of 8086 is moving in forward memory directions for executing following assembly language codes:
MOV AX, 1A2Bh;
NOT AX;
ADD AX, FFh; | 10
(CO1)
(PO1) |

After the executions of all arithmetic and logical instructions in single-step debug mode with enabling interrupt option, what values would be there in AX and Flag (CF, PF, AF, ZF, SF, OF, IF, TF, and DF) registers?

- b) Let, an assembly language program initiates with a memory segment registers as CS=1000h, DS=2000h, and SS=3000h. If the program starts with an origin of 0100h and both the data and stack segments are empty, then, derive the appropriate related register values (i.e., offset values) for each segments after execution of the assembly language codes given in Code Snippet 1.

12
(CO4)
(PO2)

```
1 org 0100h
2 .DATA
3 A db 50
4 B dw 260
5 C db "IUT OIC$"
6
7 .CODE
8 MAIN PROC
9     mov al, A
10    mov bx, B
11    push al
12    push bx
13    pop dx
14    MAIN ENDP
15 END MAIN
16 ret
```

Code Snippet 1: An Assemble Language code for Question 3.b

- c) Explain how to define a numeric constant value in an assembly language program.

3
(CO1)
(PO1)

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CSE 4511: Computer Networks

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Describe the services the network layer provides at a router using necessary diagrams. Discuss why we need to reassemble data packets at the final destination, not at each router. 7 + 3
(CO1)
(PO1)
- b) An organization is granted a class B (172.16.0.0) and a class A (10.0.0.0) network address. The administrator wants to apply the subnet mask 255.255.255.192 to create subnets from both of the given networks. As a network expert, answer the following questions for both of the networks: 10
(CO2)
(PO1)
 - i. How many subnets can be there?
 - ii. Find the number of addresses in each subnet.
 - iii. Find the first address, the last address, and the broadcast address in the first subnet.
 - iv. Find the first address, the last address, and the broadcast address in the last subnet.
- c) What are the main motivations for using variable length subnet mask (VLSM)? Consider the network with 11 subnets shown in Figure 1. Considering a class C network (192.168.10.0) design a VLSM network and give the VLSM table with an optimized worksheet representing the slash notation of each subnet address. 10
(CO2)
(PO2)

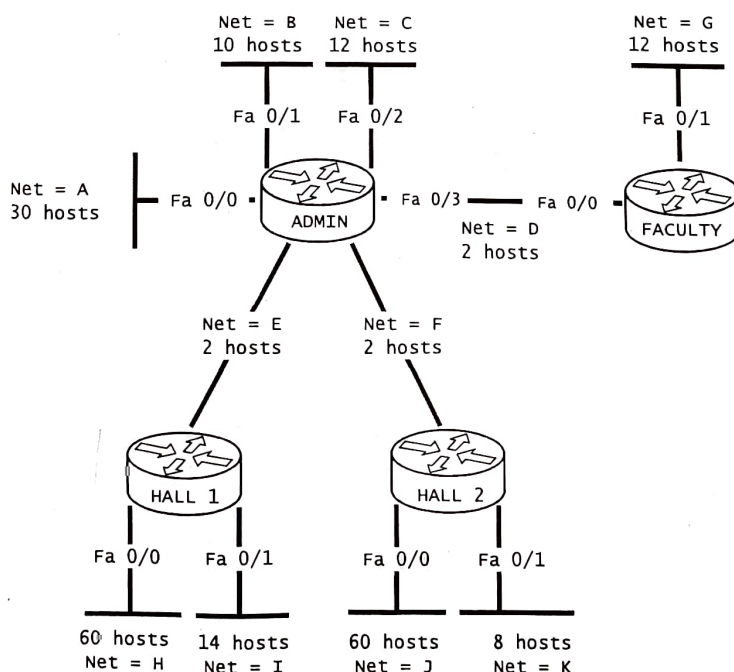


Figure 1: Network for Question 1.c)

2. a) Name the ICMP error-reporting messages. Explain the encapsulation of an ICMP packet with necessary diagrams. How does the encapsulation process of ICMP packet differ from that of an ARP packet? 10
(CO1)
(PO1)
- b) Briefly explain the host autoconfiguration feature of IPv6 addressing. Assume a host with an Ethernet address $(F5 - 00 - 00 - 00 - 00 - E2)_{16}$ has joined the network. What would be its global unicast address if the global unicast prefix of the organization is $(3A00:1111:0000)_{16}$ and the subnet identifier is $(ABCD)_{16}$? 10
(CO2)
(PO1)
3. a) What are the rationales for migrating from IPv4 to IPv6? Mention the main reasons for the delay in the adoption of IPv6 protocols. Give a comparative analysis of IPv4 and IPv6 in terms of the following: 5 + 10
(CO2)
(PO1)
- Private address
 - Reserved addresses
 - Header format
 - Resource allocation
- b) Briefly explain the working principle of link state routing. Consider the network shown in Figure 2, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance vector/table entries at node-V shared with its neighbors. How does the scenario change using a link state algorithm instead of a distance-vector algorithm? 5 + 5
(CO3)
(PO2)

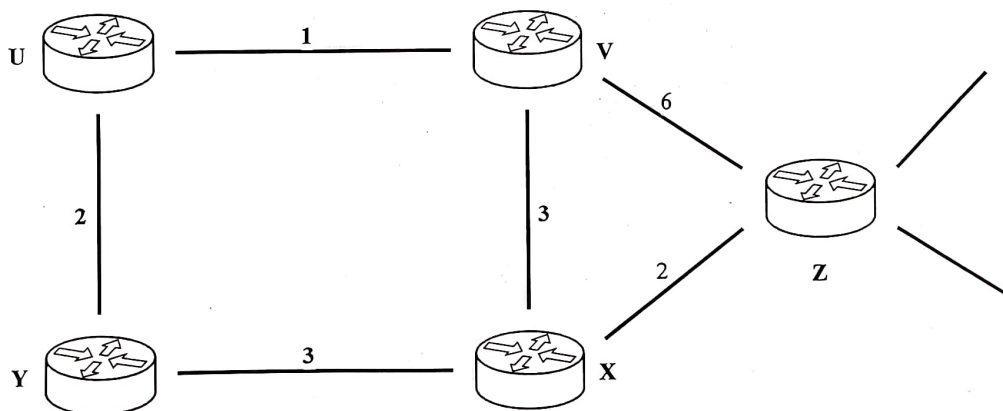


Figure 2: Network for Question 3.b

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CSE 4513: : Software Engineering and Object-Oriented Design

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Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) In a Hospital Management System (HMS), a patient can log in to the online portal to book an appointment with a doctor. Once logged in, the patient begins by searching for available doctors based on their preferred specialization and appointment date. The system displays a list of doctors along with available time slots. The patient can browse through the list and perform multiple searches until they find a suitable option. Once satisfied, the patient selects a doctor and confirms a time slot. However, the system prompts the patient to choose a different time if the selected slot becomes unavailable due to last-minute scheduling changes. After selecting the time slot, the patient proceeds to make the payment. The system initiates a payment request through the payment gateway. If the payment is successful, the booking is confirmed. In case the payment fails, the system provides an alternative option to retry the payment or select a different payment method. If the patient selects the option for insurance-based payment, the system interacts with the insurance provider to verify coverage. If the insurance verification fails, the system notifies the patient to proceed with another payment method. Once the payment is confirmed, the doctor receives a notification about the new appointment, and the selected time slot is marked as booked. Simultaneously, the notification service sends a confirmation message to the patient via SMS.

15 + 15
(CO2)
(PO2)

 - i. Using the provided scenario, create a sequence diagram that illustrates the interactions between the patient, system, payment gateway, insurance provider, doctor, and notification service throughout the appointment booking flow.
 - ii. Identify the classes in the online appointment booking use case of the Hospital Management System (HMS), along with their attributes and methods, and mention the *associations* and *multiplicities* between these classes.
2. a) A software development company is working on a healthcare application that requires a scheduling feature where user satisfaction and functionality are critical. The company has a reusable scheduling component that can be customized to fit various user requirements. They have two options —

7
(CO1)
(PO2)

Opt 1: Modify the existing scheduling component to meet the specific needs of the healthcare application, such as integrating with patient management systems while keeping the core user requirements (like appointment booking) the same.

Opt 2: Adjust the user requirements of the healthcare application to fit the existing scheduling component, allowing the company to utilize it without extensive modifications but potentially limiting some desired functionality.

Based on the scenario provided, which approach do you believe the developer company should select, and what justifies your recommendation?
- b) Formulate eight user stories and identify five non-functional requirements for the online appointment booking process within the Hospital Management System (HMS) described in Question 1.a. How the *Definition of Done* can be used to verify the completeness of these non-functional requirements?

8 + 5
(CO1)
(PO1)

- c) Based on the Burndown chart provided in Figure 1, What can be inferred after the third iteration, given that the agile team planned to release after the fourth iteration?

5
(CO1)
(PO1)

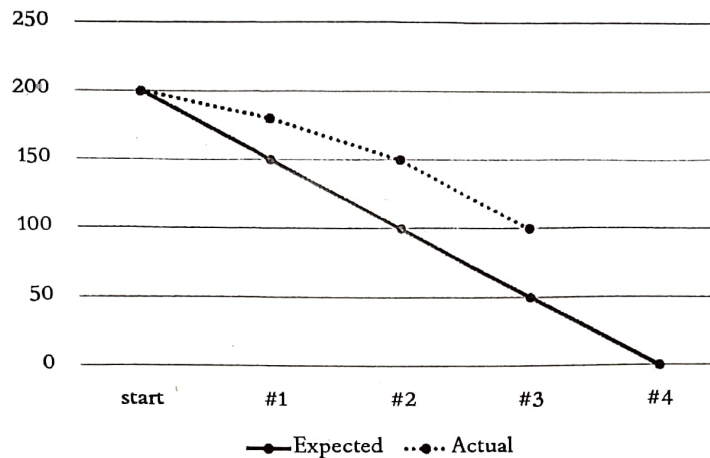


Figure 1: Burndown chart for Question 2.c

3. a) Identify at least four Internal Logical Files (ILF), three External Interface Files (EIF), two External Queries (EQ), two External Outputs (EO), and two External Inputs (EI) related to the online appointment booking process within the Hospital Management System (HMS) described in Question 1.a. Assume that all the components you identified have average (EI-10, EO-7, EQ-4, ILF-5, and EIF-4) complexity. In addition, the system requires significant performance, average end-user efficiency, moderate distributed data processing, and significant data communication. Other General system characteristics (GSCs) are incidental. Compute the adjusted function points using FPA.
- b) Referring to the diagram in Figure 2, discuss how can usability be effectively integrated? Discuss the relationship between product usability and a business's profitability.

4 + 6
(CO1)
(PO1)

3 + 7
(CO1)
(PO2)

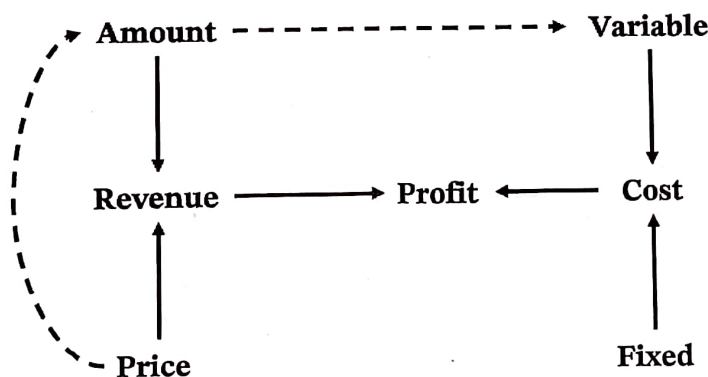


Figure 2: Diagram for Question 3.b

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DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024

FULL MARKS: 75

CSE 4551: Computer Graphics and Multimedia Systems

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suppose, in a 3D space, you have a line, L , that goes through the two points, $A(1, 0, 2)$ and $B(3, 5, 4)$. Consider that the Y -axis is facing vertically upward in the coordinate system. Now, derive the series of transformations required to rotate any cube by 45° clockwise around the line L and layout the transformation matrices in proper order. You are not allowed to use Rodrigue's formula. 10
(CO1)
(PO1)
- b) A linear kinematic chain in 2D, where Node N_m is the child of N_{m-1} and N_0 is the root node, is defined with the following information:
 - N_0 : {Offset: (0, 0), Orientation: 0° , Constraint: $[-90^\circ, 90^\circ]$ }
 - N_1 : {Offset: (5, 0), Orientation: 45° , Constraint: None}
 - N_2 : {Offset: (2, 0), Orientation: 45° , Constraint: None}
 - N_3 : {Offset: (1, 0), Orientation: -90° , Constraint: None}

An actuator is located at the origin of Node N_3 .

 - i. Based on the information, roughly sketch the kinematic chain on your script. 4
(CO1)
(PO1)
 - ii. Considering the given constraints and the offsets of each joint, identify the areas where the chain is under, over, or properly constrained based on inverse kinematics. Show the areas with the help of a drawing. Justify your answer with proper reasoning. 6
(CO2)
(PO2)
- c) Briefly explain how a graphical system benefits from instancing by leveraging scene graphs. 5
(CO1)
(PO1)
2. a) Generate the series of points for drawing a line between the two points, (3, 3) and (11, 7), using Bresenham's line drawing algorithm. Along with each of the generated points, highlight the values of the decision variable, d . 12
(CO1)
(PO1)
- b) Given you have two points, $(-1, -1)$ and $(-3, -6)$, write a pseudo code to draw a line between these two points using a Digital Differential Analyzer-based algorithm. 8
(CO1)
(PO1)
- c) How do subdivision surface and displacement mapping work together to create realistic 3D models with intricate details? Explain with the help of an appropriate example. 5
(CO1)
(PO1)

3. a) Often, while drawing diagrams for representational purposes, we draw different nodes, such as boxes, with texts in them, and then connect them with connectors, which in most cases, are curves. Some such references are shown in Figure 1.

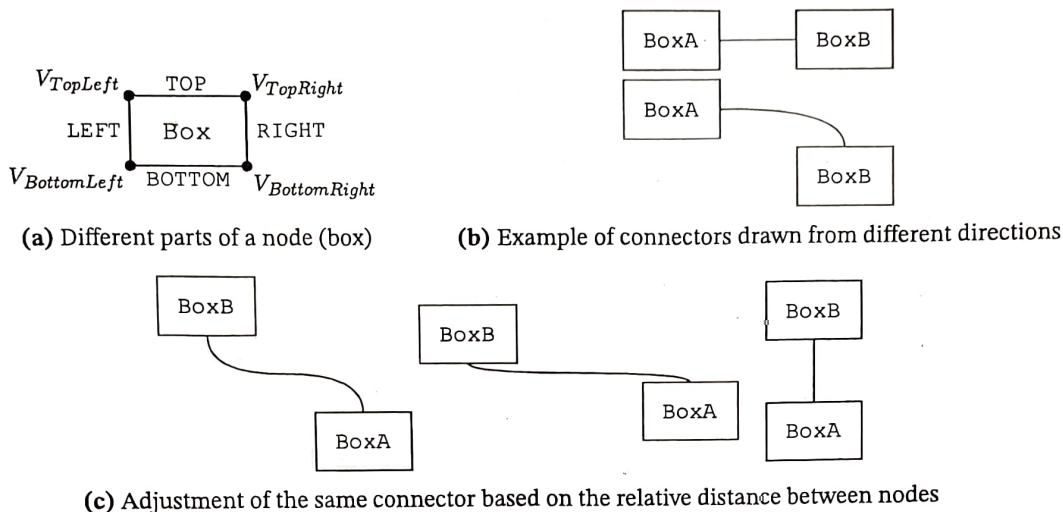


Figure 1: Curves used as connectors in diagrams for Question 3

Assume that you are given two such boxes, BoxA and BoxB. You can access the coordinates of each of the four vertices of a box ($V_{TopLeft}$, $V_{TopRight}$, $V_{BottomRight}$, and $V_{BottomLeft}$). A connector will always be drawn from the middle of an edge.

- i. Along with the aforementioned information, you are also given a direction, TOP, RIGHT, BOTTOM, or LEFT for each of the boxes. You need to design an algorithm in the form of a pseudo code to draw a connector starting from the appropriate edge of BoxA towards the appropriate direction and ending at the appropriate edge of BoxB in the appropriate direction. For instance, if the direction of the curve is given as TOP of BoxA to BOTTOM of BoxB, it would look like the examples in Figure 1(c). While deriving the algorithm, you need to satisfy the following requirements:

- The algorithm should work for all four directions properly.
- The curve should adjust its shape based on the relative distance between the boxes as shown in Figure 1(c).
- For simplicity, a curve may intersect its own box, but only in appropriate cases.

You can use any cubic curve that you think fits. You can also directly use the basis matrix of the curve of your preference by symbol. You do not need to do any specific value calculations, but need to show all the equations.

- ii. Consider that you are only given the four vertex coordinates of each box without the specific direction to draw the connector from. Design an algorithm as a pseudo code to automatically deduce the starting direction of the connector from BoxA and ending direction at BoxB so that the curve does not intersect the boxes given that the boxes do not intersect each other.

- b) Suppose, you are given a cubic Bézier curve drawn using the equation, $\vec{\gamma}(t) = G_{\text{Bézier}} B_{\text{Bézier}} T(t)$, where the symbols have usual meanings. Keeping the shape of curve intact, what will be new geometry matrix, G_{BSpline} for the same curve if drawn using the B-Spline basis, B_{BSpline} ?

12
(CO4)
(PO3)

8
(CO4)
(PO3)

5
(CO1)
(PO1)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Mid Semester Examination

Course Number: **Math 4541**

Course Title: Multivariable Calculus and Complex Variables

Winter Semester: 2023 - 2024

Full Marks: 75

Time: 1.5 Hours

There are **3 (three)** questions. Answer **all** questions. The symbols have their usual meanings. Marks of each question and corresponding CO and PO are written in the brackets.

1. a) Find z^5 for $z = -3 - i$ [5] CO1
PO1
- b) Determine the image of the quarter disk defined by the inequalities $|z| \leq 2$, $0 \leq \arg(z) \leq \pi/2$, under the mapping $w = z^3$. [10] CO1
PO1
- c) Show that the given function is not analytic at any point but is differentiable along the coordinate axes. [10] CO1
PO1

$$f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2y - y)$$

2. a) Use L'Hopital's rule to compute the following limit: [5] CO1
PO2
- $$\lim_{x \rightarrow 1 + \sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$$
- b) (i) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane. [3] CO1
PO1
- (ii) Find the harmonic conjugate function of u . [7]
- c) Evaluate $\oint_C y^2 dx - x^2 dy$, where C is the closed curve shown in Figure 2(c). [10] CO1
PO1

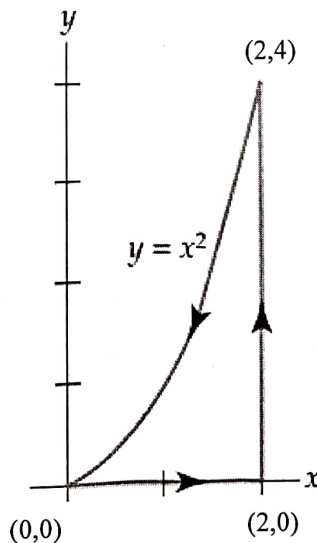


Fig. 2(c)

3. a) Find the slopes of the surface

[5] CO2
PO1

$$f(x, y) = 1 - (x - 1)^2 - (y - 2)^2$$

at the point (1, 2) in the x -direction and in the y -direction.

b)

Consider the function given by $f(x, y) = \sqrt{16 - 4x^2 - y^2}$

[10] CO2
PO1

(i) Find the domain and range of the function f .

(ii) Describe the graph of f .

c)

Using Cauchy's Integral formula evaluate $\oint_C \frac{z}{z^2 + 9} dz$

[10] CO2
PO1

where C is the circle $|z - 2i| = 4$.

Mid-Term Examination Questions

Islamic University of Technology (IUT) has three campuses across different cities: Gazipur, Chattogram, and Rajshahi. Each campus has a different number of students, departments, and network requirements. IUT requires a robust network infrastructure to connect these geographically separated offices while ensuring efficient communication, security, and scalability.

As a network engineer, you are tasked with designing and configuring networks to connect these three campuses to allow efficient communication.

Campus Overview

The campuses consist of multiple buildings, housing departments and offices:

1. Gazipur Campus

- i. Computer Science & Engineering (CSE)
- ii. Electrical & Electronic Engineering (EEE)
- iii. ICT Center
- iv. Administration Department

2. Chattogram Campus

- i. Civil Engineering (CEE)
- ii. Mechanical & Production Engineering (MPE)

3. Rajshahi Campus

- i. Technical & Vocational Education (TVE)
- ii. Business & Technology Management (BTM)

Network Requirements

- All devices (computers, etc.) should be configured dynamically (i.e., IP address, subnet mask, default gateway, and DNS server should be assigned automatically).
- The ICT Center should host all servers of IUT.
- Each department and office must be isolated from each other in terms of broadcast traffic.
- Implement appropriate routing protocols to allow communication where necessary.
- The Administration Department should monitor all the traffic to and from the ICT center.
- The University maintains a server for sensitive data (financial, student records, etc.). Only the Administration should have access to this server.
- Each switch port should be configured to allow only two (2) devices. If a port violation occurs, the port should be shut down for the Administration departments. For others, the switch port should simply drop packets in case of a violation.
- The University has purchased three (3) public IP addresses, $203.16.x.x+1/24$, $203.16.x.x+2/24$ and $203.16.x.x+3/24$, where x represents the last two digits of your student ID.

Each of them will be used separately for each campus. These IPs will be used for internet access. All internal IP addresses must be mapped to these public addresses for external connectivity.

Task

Your task is to implement a functional, efficient, and secure network for IUT using Cisco Packet Tracer that meets all of the specified requirements and allows for future growth and expansions.

- * Assume a suitable network address and subnet mask for your solution.
- * Assign the IP addresses manually where necessary.
- * Connect at least one device for each department.